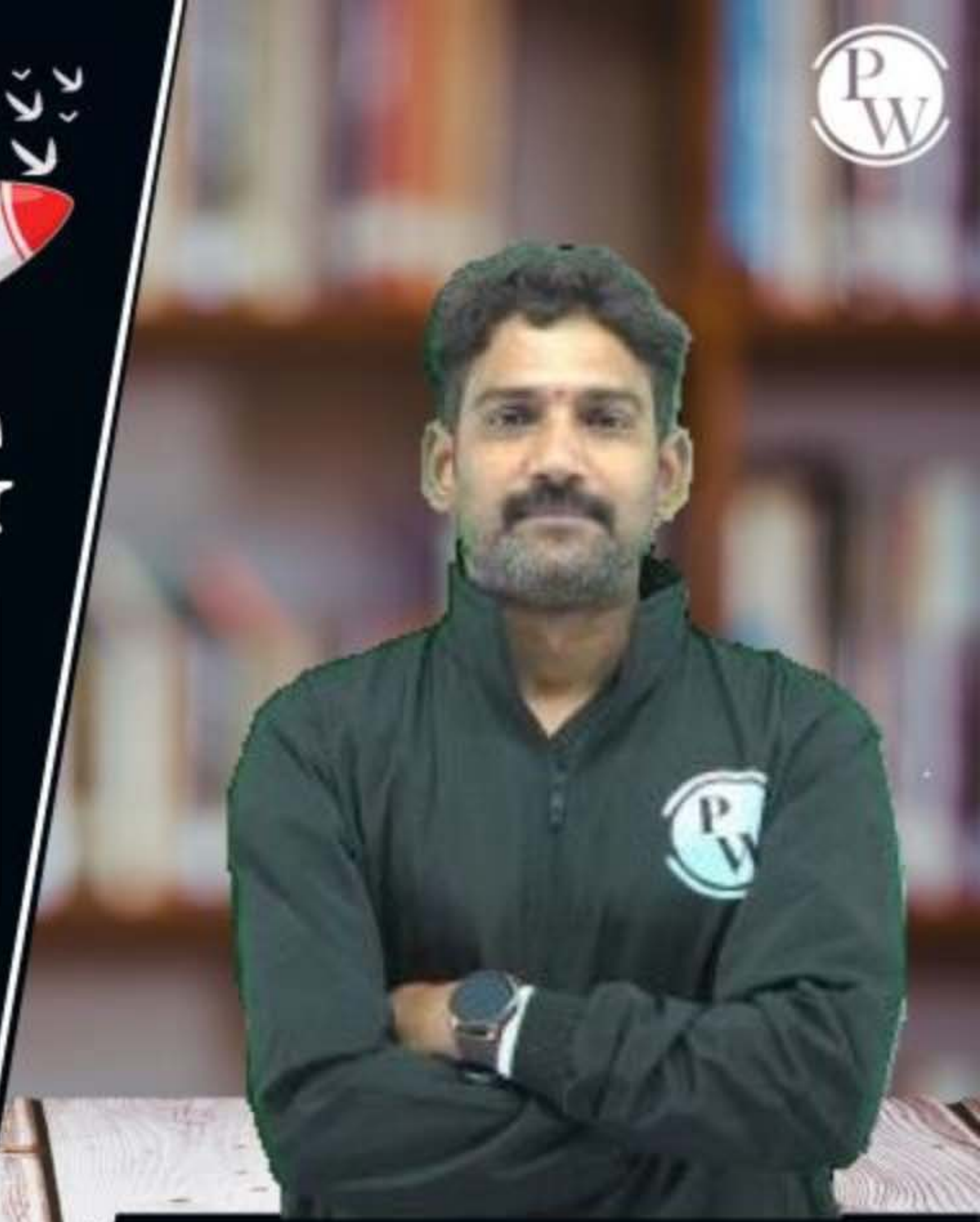
CS & IT ENGINEERING

Data Structure & Programming

Hash Tables

Lecture No.- 01



By- Satya sir

Recap of Previous Lecture

- Stack Implementation
  - Using Queues
- Queue Implementation
  - Using Stacks

Topics to be Covered

NOTE: Tomorrow Extra class @ 7AM

- Hashing
- Hash Function
- Hash Table
- Collission Resolution Techniques
  - Chaining Method
  - Linear Probing

Topic : Hash Tables - 1

Insertion

- Array → Random Insertion
  - End $O(1)$
  - middle/Beginning $O(n)$
- Stack → At the top only
  - → Array ⇒ Insertion at End $O(1)$
  - → Linked List ⇒ Insertion at Beginning $O(1)$
  - → Queue
    - Push costly $O(n)$
    - Pop costly $O(1)$
- Linked List (SLL) →
  - Beginning $O(1)$
  - middle $O(n)$
  - End $O(n)$
- Queues →
  - → Rear End only (Simple Q, CQ)
  - → Rear | front (DQ)
  - Array — Insertion at End $O(1)$
  - Linked List ⇒ $O(1)$ | $O(n)$
  - Stack
    - Enqueue Costly $O(n)$
    - Dequeue Costly $O(1)$

Hashing $\cong$ mapping (Matching)

- Hashing is the Process of mapping key to Value.

  Key : The Number that need to be accessed/Stored in Memory.

  Value : Location | Slot | Bucket | Block | frame where, key is stored.

- Memory where indices are maintained is Said to be Hash Table.

- Hashing is done Using Pre-defined mathematical Hash Function.

  Ex: $h(k) = k \% n$ // Primary Hash Function

  $h(k) = (2k+3) \% n$

  $h(k) = (k^2+7) \% n$

  n = Size of Hash Table (No. of slots)

Topic : Hash Tables - 1

- Let Hash Table of size 10 [numbered from 0 to 9]

The keys : 27, 13, 7, 3, 5, 35, 44

$$h(K) = (K+3) \% n$$

$h(27) = (27+3) \% 10 = 30 \% 10 = 0$

$h(13) = (13+3) \% 10 = 16 \% 10 = 6$

$h(7) = (7+3) \% 10 = 10 \% 10 = \underline{0}$?

$h(3) = (3+3) \% 10 = 6 \% 10 = 6$?

$h(5) = (5+3) \% 10 = 8 \% 10 = 8$

$h(35) = 38 \% 10 = 8$?

$h(44) = 47 \% 10 = 7$

Hash Table

0	27
1	
2	
3	
4	
5	
6	13
7	44
8	5
9	

Value (7) == Value (27)

Value (13) == Value (3)

Collission

When Two or more keys resolve to same value in the Hash Table, it is said to be Collission

Collission Resolution Techniques

2 Types

1. Open Hashing method (or) Seperate chaining method
2. closed Hashing method (or) open Addressing Method
   - It has 3 Sub Types of Resolution Techniques
     1) Linear Probing
     2) Quadratic Probing
     3) Double Hashing

Seperate Chaining method

- In this method, all keys that resolve to same value, are organized Using singly Linked List as a chain of keys.

Ex: Let, Hash Table size = 11   Hash Function, $h(K) = K \% 11$

keys = 23, 37, 48, 67, 15, 92, 78, 50, 30, 28, 80

$h(23) = 23 \% 11 = 1$

$h(37) = 37 \% 11 = 4$

$h(48) = 4$

$h(67) = 1$

$h(15) = 4$

$h(92) = 4$

$h(78) = 1$

$h(50) = 6$

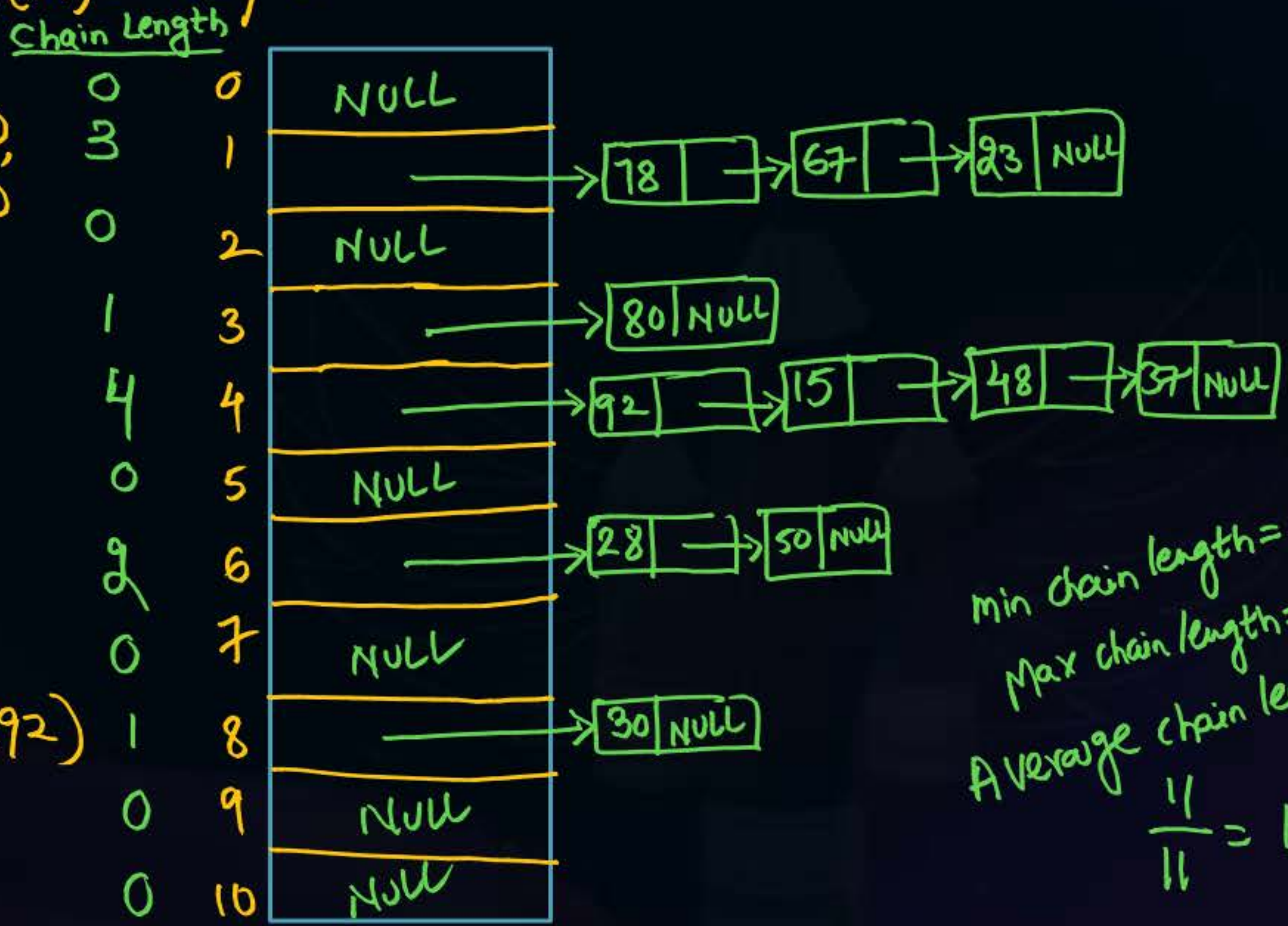
$h(30) = 8$

$h(28) = 6$

$h(80) = 3$

$h(23) == h(67) == h(78)$

$h(37) == h(15) == h(48) == h(92)$



min chain length = 0

Max chain length = 4

Average chain length

$\frac{11}{11} = 1$

Drawbacks

1. Requires Extra Memory Space
2. Sequential access ⇒ $O(n)$

Open addressing Techniques

— These are Sub-classified into 3 Techniques :

1) Linear Probing
2) Quadratic Probing
3) Double Hashing

Linear Probing : Hash Function, $h(k, i) = (k+i) \% n$ k = key, n = Table size, i = Probe Number.

→ $i=0$ for Each key Initially.

→ When Collision Occurs, then only $i = i+1$.

→ When $i=0$, i.e in the first Probe, $h(k,0) = (k+0) \% n$

$\boxed{h(k,0) = k \% n}$ Primary Hash Function.

Topic : Hash Tables - 1

$H.F,\ h(k,i) = (k+i)\%n$

Example

Consider Hash Table size=9, $h(k) = k\%9$, Linear Probing

Keys = 29, 37, 60, 49, 24, 40, 19, 15

$h(29,0) = 29\%9 = 2$

$h(37,0) = 37\%9 = 1$

$h(60,0) = 60\%9 = 6$

$h(49,0) = 49\%9 = 4$

2 Probes $\begin{cases} h(24,0) = 24\%9 = \underline{6} \text{ collission} \\ h(24,1) = (24+1)\%9 = 25\%9 = 7 \end{cases}$

$h(40,0) = 40\%9 = 4$ Collission

$h(40,1) = (40+1)\%9 = 5$

$h(19) = 19\%9 = 1$ collission

$h(19,1) = (19+1)\%9 = 20\%9 = 2$ Collission

$h(19,2) = (19+2)\%9 = 21\%9 = 3$

$h(15,0) = 15\%9 = 6$ Collission

$h(15,1) = (15+1)\%9 = 7$ Collission

$h(15,2) = (15+2)\%9 = 8$

3 Probes

Hash Table

0	
1	37
2	29
3	19
4	49
5	40
6	60
7	24
8	15

15

24

2 mins Summary

Drawback of Linear Probing :

- Consecutive slots form as a group and increases Number of Probes for Each key.

  It is called as Primary clustering issue.

- Hashing, Hash Function, Hash Table
- Collission Resolution Techniques
  - seperate chaining method
  - Linear Probing

To be Contd ...

THANK - YOU